In-browser FTMS simulations: from isotopic calculations to top-down mass spectra

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Introduction

- FTMS offers exceptional resolution and accuracy, but modern applications especially in top-down proteomics and complex mixture analysis - generate highly complex, data-rich spectra that challenge current analysis tools.
- Accurate benchmarking of MS data processing tools requires access to "gold-standard" datasets that reflect real experimental complexity across instruments and analyte types.
- In-silico simulations of FTMS data can aid both experimental design and algorithm development by providing known ground-truth spectra under defined conditions.
- Existing simulation tools often lack accessibility, flexibility, or sufficient data security, limiting their use in educational, industrial, and regulated biopharmaceutical settings.
- To address these gaps, we developed the FTMS Simulator (web and desktop versions), launch the ProteoGold dataset initiative, and propose it as a platform for benchmarking, education, and development in top-down MS analysis, and other applications, e.g., complex mixture analysis.
- FTMS simulator is a software tool to accurately simulate isotopic envelopes and mass spectra:
  - to generate "gold-standard" datasets for the development and validation of data analysis software tools across diverse FTMS applications, e.g., top-down proteomics & complex mixture analysis
  - to verify the experimental results and hypotheses, perform data quality control
  - to teach the FTMS subject by demonstrating realistic transients and corresponding mass spectra
  - to search the experimental data with a like-for-like simulated data using similarity score's ranking
  - to design FTMS experiments by selecting optimum settings and FT data processing parameters

Workflow for FTMS spectra generation

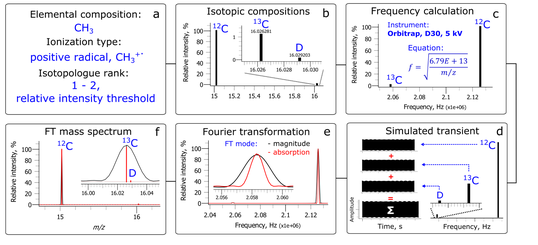

Figure 1. Schematic diagram of the workflow of Fourier transform (FT) mass spectra simulation for user-defined chemical compounds via in silico time-domain signal (transient) generation for ICR or Orbitrap FTMS instruments. (a) Definition of molecular formulas of a compound of interest, (b) calculation of a complete isotopic distribution on a m/z scale, (c) conversion of the calculated isotopic distribution from a m/z scale into a frequency scale, (d) simulation of transients composed of sinusoidal components that correspond to individual isotopologues (m/z values and abundances), (e) FT processing of transients, and (f) visual representation of transients, isotopic distributions, and mass (or frequency) spectra. (JASMS, 2020, 1927; DOI: 10.2021/jasms.0c00190)

Workflow for TD mass spectra generation

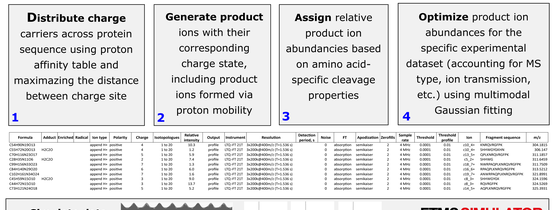

Figure 2. Schematic diagram of the workflow of broadband top-down mass spectrum simulation. Transients are simulated individually for each product ion with the corresponding abundancies, summed up together and processed using the user defined FT settings. The noise, decay and phase distortions may be additionaly applied.

In-browser FTMS Simulator: www.peakbypeak.com

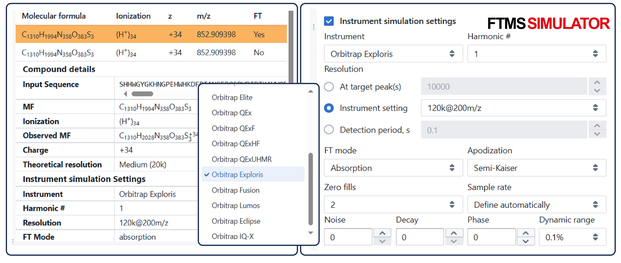

Figure 3. In-browser FTMS Simulator settings. **Compound definition:** elemental composition, amino acid sequence, isotopes relative abundance definition, modifications, mass (m/z) or frequency value. **Ion (charged compound) definition:** Charge carrier: electron, H+, K+, Na+, Cs+, I-, HCOO-; Ionization: positive, negative, or neutral; Instruments: ICR/MRMS, Orbitraps; Charge state: selected or range; Isotopologues: how many and which ones; **FT processing settings:** Resolution: at target peak, instrument setting, transient length; FT mode: absorption or magnitude; Apodization window: none, full (Kaiser), half (semi Kaiser); Number of zero fills: 0, 1, 2, or 3; Sampling rate: 1, 2, 4, 6 MHz, or any.

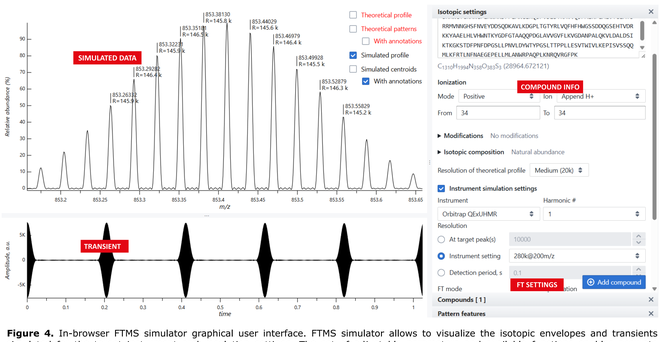

Figure 4. In-browser FTMS simulator graphical user interface. FTMS simulator allows to visualize the isotopic envelopes and transients simulated for the target instrument and resolution settings. The set of adjustable parameters and available functions enables accurate simulation and representation of FTMS data of practically any complexity. Carbonic anhydrase was simulated for Orbitrap QEx instrument.

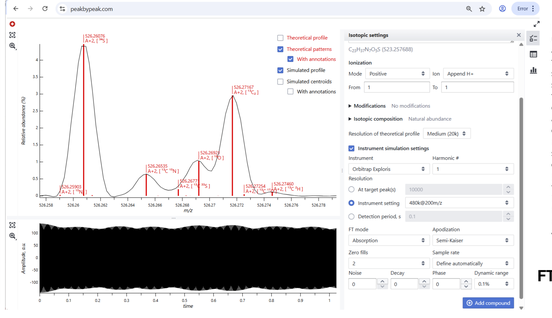

Figure 5. In-browser FTMS simulator graphical user interface. The tool allows to overlay the theoretical pattern and simulated profile mass spectrum. It allows user to find optimal experimental settings to achieve target objectives, e.g. to resolve isotopic fine structure of a target compound. MRFA peptide was simulated for Orbitrap Exploris instrument at the highest available resolution settings 480k@200m/z.

FTMS SIMULATOR

"Gold-standard" top-down mass spectra generation: Comparing experimental and simulated MS/MS mass spectra of carbonic anhydrase II

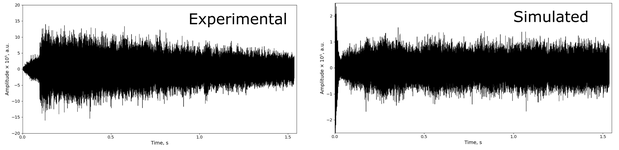

Figure 6. Left panel: TD ETD MS/MS experimental data acquired on a 21 T FT-ICR MS (NHMFL) (Weisbrod *et al.*, JASMS, 2017) and right panel: simulated (a/y/c/z ions) of carbonic anhydrase II (CA2, MW 29 kDa, precursor charge state 34+) via transient generation with FTMS Simulator. .

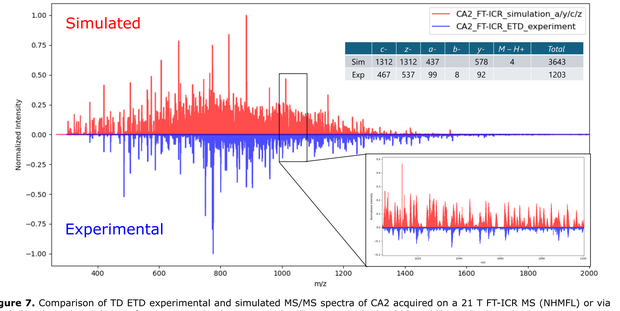

Figure 7. Comparison of TD ETD experimental and simulated MS/MS spectra of CA2 acquired on a 21 T FT-ICR MS (NHMFL) or via FTMS Simulator (*a/y/c/z* ions for precursor 34+), respectively. The zoom-in (*m/z* 1000–1100) emphasizes the isotopic structure agreement and demonstrates similar spectral complexity.

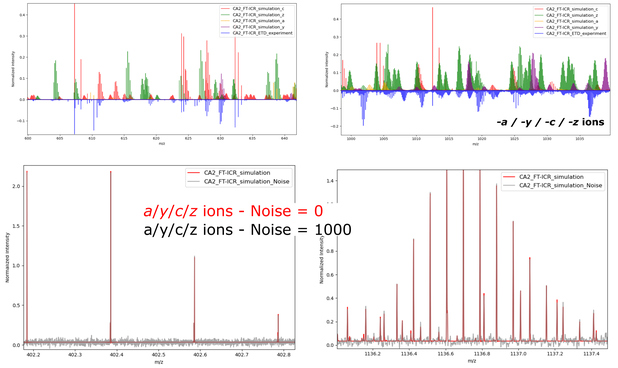

Figure 8. STop panels: the simulated MS/MS spectrum including a/y/c/z ion series in different colors vs TD ETD experimental data. Bottom panels: zoomed-in regions (left: low m/z, right: high m/z) comparing noise-free and noise-added simulations for spectra containing a/y/c/z ions.

Simulated *c/z* mass spectrum

Sequence coverage 93.8 %
c-ions: 663; *z*-ions: 588

Experimental mass spectrum

Sequence coverage 90.0 %
c-ions: 351; *z*-ions: 361

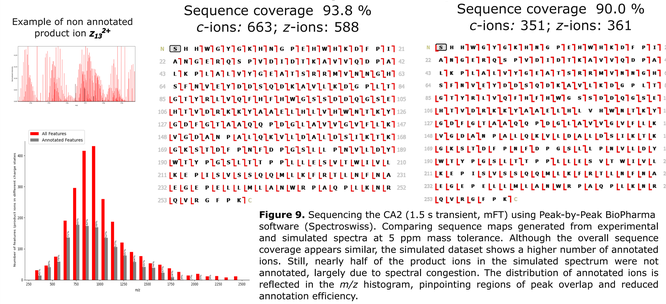

Figure 9. Sequencing the CA2 (1.5 s transient, mFT) using Peak-by-Peak BioPharma software (Spectroswiss). Comparing sequence maps generated from experimental and simulated spectra at 5 ppm mass tolerance. Although the overall sequence coverage appears similar, the simulated dataset shows a higher number of annotated ions. Still, nearly half of the product ions in the simulated spectrum were not annotated, largely due to spectral congestion. The distribution of annotated ions is reflected in the *m/z* histogram, pinpointing regions of peak overlap and reduced annotation efficiency.

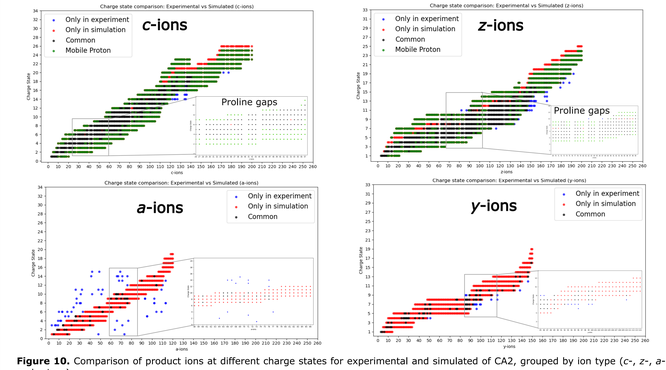

Figure 10. Comparison of product ions at different charge states for experimental and simulated of CA2, grouped by ion type (*c-*, *z-*, *a-*, and *y*-ions).

Conclusions

- FTMS Simulator is now publicly available online at www.peakbypeak.com, offering real-time, in-browser mass spectrometry simulations with secure isotopic calculations and server-supported transient generation.
- A more powerful desktop version of the FTMS Simulator enables complex simulations, such as high-resolution top-down MS spectra of CA2 and other large proteins.
- This initiative launches the ProteoGold project, which aims to build an open database of FTMS spectra and transients (10–150 kDa proteins) across instruments and experimental setups.
- Initial datasets for CA2 are available in open formats, supporting benchmarking and development of bioinformatics tools for top-down proteomics.
- Top-down FTMS spectra are highly complex, and this work highlights the need for robust tools and reference data to improve processing and interpretation accuracy.

Acknowledgements

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